CHM 1151 | Section 001 | Villanova University

General Chemistry I – Syllabus 1.0

Department of Chemistry and Biochemistry

Spring 2025 | CRN 35967

At Villanova, we acknowledge that our institution sits upon the unceded land of the Lenni-Lenape peoples.[1] We recognize that acknowledgment is only a first step; our enduring commitment to an anti-racist campus involves a process of learning about and working for the dismantling of all ongoing legacies of oppression. Therefore, this course has a tribute (on Blackboard) to the Lenni-Lenape peoples and others whose important legacies have been overshadowed by time. *To be responsible scientists in the future, we must certainly learn about, understand, and be respectful of our past*.

General Information

Student Help Hours: *By student request* • Via Zoom, email, or in person • *Dr. Katz is here to help!*

Class Location and Times: Mendel 115 • Tues and Thurs 3:00 pm – 4:40 pm • Attendance is mandatory

Professor: Dr. Stephanie Katz Linkmeyer (she/her) • Associate Teaching Professor • Department of Chemistry and Biochemistry

Professor Contact: Stephanie.A.Katz@Villanova.edu • 954-270-8864 (call/text) • Monday - Saturday • 9:00 am – 10:00 pm

Course Description

CHM 1151, General Chemistry I, is a survey of fundamental chemical concepts and models. Theory and practice are used to emphasize *critical thinking and problem-solving skills* with respect to chemistry. As chemistry is an experiential discipline, concepts build on one another and *deliberate practice* is required on the part of the student to ensure understanding and mastery of the topics introduced by the instructor. Topics to be covered are described below.

Required Course Materials

Open Stax, Chemistry, Atoms First 2e, (OSAF), **$0.00 provided via Blackboard (Bb)**. Print copy available for low cost.

Achieve, Macmillan Learning, (~$40), Homework (HW) platform, **provided via Bb**.

A free, preview copy is available on the first day of class. Students who are not in the All-Access Program must pay within the first two of weeks of class. Important: if this creates a financial burden, please inform Dr. Katz!

Webcam, external or internal, connected to the computer used for weekly online testing. (~$20)

Most computers have a webcam; please make sure it is functional.

Scientific Calculator *must be capable of doing exponents and logarithms*. **Students may NOT use a phone calculator for testing.**

Correlating Chapters

The core ideas described on page 3 can be found in the following chapters in the **OSAF** text. Additional resources are provided:

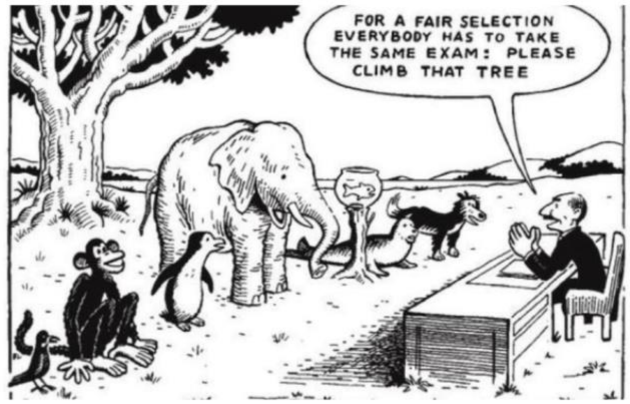
TOPIC	OSAF	Additional resources on Bb
I. Essential Ideas: Mathematical Treatment of Measurements	1	• Podcasts • Learning Checklists • Study Guides for Quizzes and Exams • Sustainability Lessons • Chemistry Journals
II. Atoms, Molecules, and Ions	2	
III. Electronic Structure and Periodic Properties of Elements	3	
IV. Chemical Bonding and Molecular Geometry; IMFs	4; 10.1	
V. Advanced Theories of Covalent Bonding	5	
VI. Composition of Substances and Solutions	6	
VII. Stoichiometry of Chemical Reactions	7	
VIII. Gases	8	
IX. Energy Basics, Calorimetry, Enthalpy, and Hess's Law	9	
X. Liquids and Solids	10	

Professor Availability

This semester, Dr. Katz will be most likely be in Mendel 214 B, directly across from the lecture classroom, Mendel 213, on Tuesdays and Thursdays after class ends. Students should feel free to stop in any time. If a more formal meeting is preferred, please email Dr. Katz to arrange a time that works. Students who email to make an appointment will likely receive a reply within an hour, if not a few minutes. ***But if a reply hasn't been received in 24 hours, it's likely there is an email issue*** (check outbox and spam folders). Another option for contacting Dr. Katz is to come early or stay late after class and set up a meeting that way. *Please reach out when struggling*. **Note: Dr. Katz may not reply to emails/texts on Sundays, or between 10:00 pm and 9:00 am, Monday – Saturday.**

[1] Please visit https://nlltribe.com/land-acknowledgement-before-import/ or Syllabus and Professor Bb page for the land acknowledgment preferred by the Lenape peoples.

Commitment to Justice, Equity, Diversity, and Inclusion (JEDI)



Source: https://marquetteeducator.wordpress.com/2012/07/12/climbthattree/

It is the highest priority that this course foster a supportive, effective learning environment for all students based on empathy and mutual respect, regardless of whether or not the curriculum directly addresses issues related to justice, equity, diversity, and inclusion (JEDI). It is similarly important that no one in the course experiences discomfort due to any area of personal identity. In this regard, students from all backgrounds and identities will be honored, valued, and viewed as a resource, strength, and benefit to this course in the hope that this creates an environment where everyone feels they can be successful. ***To create this class ethos, we'll start with these commitments***: (1) All people in the class will engage in respectful discussion, avoiding language that is harmful to another person; (2) If academic performances in this class are being impacted by non-academic experiences in the class, please talk with someone*; (3) If any person has pronouns that they would like their classmates or professor to use, please feel comfortable to share them; (4) If something was said in class (by anyone, especially Dr. Katz) that caused discomfort, please let someone* know about it (see below); and 5) please contact Dr. Katz with any suggestions to improve the quality of the course materials, especially in the context of JEDI. Suggestions are encouraged and appreciated. Read about the University's commitment to JEDI.[2]

Finally, and **most importantly**, at Villanova, we understand that some students are uncomfortable going directly to their professors to discuss microaggressions (or other instances of concern) that occur in the classroom, so we've created an avenue for students to share these concerns with a third party. This type of reporting (**Climate Concern form***) is confidential (or anonymous if preferred) and will be addressed with the person about whom the report is filed. These types of reports are important for *personal growth* as we all strive to become more understanding of each other's comforts and boundaries. A link to the form can be found on the Syllabus and Professor page on Bb, as well as ODEI's website.[3] [NOTE: on this syllabus, words highlighted in blue are ***pages on the left menu on Bb***.]

General Expectations

- *Expectations* – Course expectations for this class will be co-constructed during the first class meeting. Students and the instructor will collaborate to establish mutually agreeable course expectations and sign a course contract. Student groups will similarly be established. A small part of each student's grade will be determined based upon the work done with partners.
- *Before Class Preparation* – As indicated below, please choose a preferred method of ***content retrieval***. This is essential for success in this course. Prior to class is also a good time to create Chemistry Journal entries for the chapter the class is about to cover.
- *During Class* – ***Guided lecture notes*** can be found on Bb on the Course Content page by chapter. Please bring either an **electronic** or **print version** to class every day. Please be aware that our biweekly 3:00 pm class, *including students' presence*, will be recorded daily. A *non-synchronous* recording of the lesson, projected notes, and all words spoken in the room will be available *by the next morning* on the Classroom Lecture Capture (CLC) page on Bb listed by date. White board notes will also be uploaded, as these are not completely viewable from the recording. Students who miss class should please view the recording of class (and view the *filled-in guided notes on the* Course Content *page*).
- *Attendance* – Because robust learning requires active participation, ***attendance is required and expected***. Accordingly, a small percentage of the course grade is allocated toward attendance and participation. Expectations related to this are participating during ABCD-card questions, assisting with group assignments, and attending at least 90% of our T, Th lectures. This allows each student two unexcused absences without penalty *plus one personal day allowed by the University* for a total of 3 *unexcused* absences. **However, students who email to report they're unable to attend class will AUTOMATICALLY be excused from class! Students do not have to indicate WHY they'll be missing class, just THAT they'll be missing class – this is how to get an excused absence. This policy is above the 3 *unexcused* absences (i.e., no show, no email = unexcused/personal day).** As stated above, attendance is required and expected; CLC is provided for extenuating circumstances (e.g., excused absence, quarantine, athletic competition, etc.) but isn't meant to turn the course into an asynchronous class for any student.
- *After Class* – Each chapter has homework (HW) associated with the lessons we complete in class. These HW assignments can be found on the Achieve page on Bb. Due dates are 5:00 pm on the date of the quiz for each chapter. These are ***soft deadlines***, which means that they are ***strong suggestions for optimum learning but there is no penalty for missing these deadlines***. Of note: students don't earn credit for incomplete or incorrect HW assignments.

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[2] ODEI's webpage: https://www1.villanova.edu/villanova/provost/diversity/diversitystatement.html
[3] Report Climate Concern: https://www1.villanova.edu/university/diversity-inclusion/report-climate-concern.html

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Flipped Classroom

Each flipped classroom is a little bit different, but the overarching concept is similar: what students traditionally do in class (learn content) is done outside the classroom, and what students traditionally do outside of classroom (apply content) is done in the classroom. Therefore, students are responsible for **CONTENT RETRIEVAL** – that is, students must **watch podcasts, access web content, or read the textbook** to get a first glimpse at the content for each topic. Students should do this BEFORE attending class to be best prepared for class. In class, the content will be applied to new situations: that is, students will work problems and do activities that will help with learning the content. Students will find the best success in the flipped classroom if they do regular content retrieval.

Graded Assessments (Syllabus 1.0)

Link2Learn Assignments	7.69 %	50 points	Bonus points available
Participation/attendance	5.38 %	35 points	Includes participation, attendance, polling, etc.
Homework (HW) (10 pts each)	18.46 %	120 points	12 HWs for 10 chapters (+1 XC assignment)
Sustainability (10 points each)	12.31 %	80 points	7 group activities and one self-guided lesson
Chemistry journal (5 points each)	7.69 %	50 points	10 chapters: vocabulary and equations list
Online quizzes (10 points each)	17.69 %	115 points	11 quizzes; +5 pt MUST "quiz"; *2 quizzes may be retaken*
Online exams (50 points each)	3 @ 7.69 %	150 points	3 exams; *Exam 1 or Exam 2 may be retaken*
Online final exam (50 points)	7.69 %	50 points	Tues, May 6 (4:00 pm – 6:30 pm) online (**no retake**)
	~100 %	650 points	611 points needed for an A (94%)

Syllabus 1.0 vs Syllabus 2.0

Because each student in this course arrives with varied interests, backgrounds, and schedules, this course syllabus has some flexibility built into it. Students may **opt to use Syllabus 2.0** in lieu of Syllabus 1.0. The difference between the two syllabi is the assignment Link2Learn (L2L, described below; also, full assignment description can be found on Bb). **Syllabus 2.0 students do not have to complete any L2L assignments**. All other assignments, due dates, and assessments are the same. The main difference is that students who choose Syllabus 2.0 (no L2L required) will have a 600-point semester and students who remain with Syllabus 1.0 will have a 650-point semester. View the table below to understand what each point system becomes in percentages. Once chosen, students are not permitted to change their mind later in the semester. **Please sign and upload with signature to Bb (page 9)**, and that will be a contract that indicates each students' understanding of which syllabus is the one they will follow. Graded assignments are described below.

Graded Assessments	Syllabus 1.0		Syllabus 2.0		Type
Link2Learn Assignments	7.69 %	50 points	----	-----	soft
Participation/attendance	5.38 %	35 points	35 points	5.83 %	n/a
Homework (HW) (10 pts each)	18.46 %	120 points	120 points	20.00 %	soft
Sustainability (10 points each)	12.31 %	80 points	80 points	13.33 %	soft
Chemistry journal (5 points each)	7.69 %	50 points	50 points	8.33 %	soft
Online quizzes (10 points each)	17.69 %	115 points	115 points	19.17 %	HARD
Online exams (50 points each)	3 @ 7.69 %	150 points	150 points	3 @ 8.33 % = 25 %	HARD
Online final exam (50 points)	7.69 %	50 points	50 points	8.33 %	HARD
611 points needed for an A (94%)	100 %	650 points	600 points	564 points needed for an A	

Link2Learn: 5 times, for 10 pts each, write (a) the *title* of a science- or sustainability-related or adjacent article (1 pt); (b) where it was found (1 pt) (this is not the citation; Dr. Katz is curious where/how students find news – Tik Tok, X, Google); (c) "Decolonize the literature" (2 pts) – i.e., report the author's (i) current home country, (ii) gender, (iii) race, (iv) ethnicity, and (v) education; d) write a brief summary of the article (3 pts) – that is, summarize what was learned in approximately 1 – 2 paragraphs; and (e) provide a complete MLA/APA-type citation. ***Bonus points*** available in this assignment. (Full assignment description on page 9 below).

Participation: Working with partners during class activities, attending most classes (or *emailing in advance about not attending*), and participating in polls, surveys, and the like will earn these points. *Please don't come to class when sick*; missed lessons can be viewed later on the CLC page on Bb. Students are allowed 1 personal day (per University policy) plus 2 unexcused absences without penalty (per Dr. Katz policy – no show, no email = unexcused). However, emailing to indicate absence (*not why, just that*), **will automatically result in an EXCUSED absence** (per Dr. Katz policy – email = excused absence). This policy is not meant to turn the course into an asynchronous course; it's meant to provide hassle-free flexibility to students who need it.

HW: Complete (with correct answers) chapter HW (**Achieve**) for Chapters 1 – 10. Due dates are strong suggestions. *No penalty for missing due dates*. 10 points for each set completed correctly. Partial points for sets partially correct. These are content-based questions, most with math. These may be completed these with others and students may ask for help from anyone. Since HW is integrated with Bb, the earned grades will show up in My Grades in Bb. Experience demonstrates that there is ***a brief lag time*** between when grades are earned in **Achieve** and when they show up in Bb. *Patience is greatly appreciated.* Periodically, a student may be unable to answer a HW problem correctly ***after 3 reasonable tries***, but don't click 'Give up' – instead, contact Dr. Katz (or inquire in class). Clicking the "give up" button means giving up the opportunity to answer the question correctly (and earn points for doing so). *Persistence and patience are required for success! This semester, Achieve comes with an AI tutor. Stay tuned for more about this!*

Sustainability: These are in-class, group assignments that pop up naturally during the chemistry content. Most of these activities will be done with partners and submitted as a group on Bb (Sustainability). 10 points for *completion;* students are encouraged to view the answer keys to learn correct answers. Concepts will be tested in exams and quizzes. At least ⅓ of the lessons this semester are dedicated to Sustainability. One of the assignments included in this category is a self-guided assignment.

Chemistry journal: Students will be creating *electronic pages* (e.g., pdf or docx) of a Chemistry Journal to learn chemistry *vocabulary* and *equations* in the course. Submitting each one will earn a student 5 points. Students are permitted to use the Ch 1 – Ch 10 Key Terms and Equations document to create their own (cut and paste allowed) and submit on Bb for credit.

Quizzes: Each chapter will have a *10-point* chapter quiz that students take online between Thursday at 5:00 pm and Monday at 5:00 pm (Ch 3 and 4 have 2 quizzes). Students may retake TWO quizzes *one time*, without penalty, *after getting help on the material they didn't understand*. *A missed quiz* may be taken as a retake. Quizzes are taken on Bb (Quizzes and Exams) and monitored by Respondus Lockdown Browser (LdB) with Monitor; *they are not to be taken with others*. Students have 2 hours to take each quiz unless they have legally-allowed accommodations. The MUST (*Math Up Skills Test)* quiz (30 min, *bring calculator*) is taken only once on the first day of the semester for 5 completion points. The score earned on the MUST quiz (Achieve) will help Dr. Katz guide students and *will not count toward students' final grades*. This is the only 30-minute quiz, and the only quiz students will take in the classroom this semester. Quizzes have HARD deadlines; that is, if quiz is not taken in the allocated time, student will receive a 0 on the quiz.

Exams – Multi-chapter exams will be given three times during the semester for 50 points. Students may retake ONE exam (either Exam 1 or 2), *one time,* without penalty, *after getting help on the material they didn't understand*. *One missed exam* may be taken as a retake. Exams are taken on Bb (Quizzes and Exams) and monitored by Respondus LdB; *they are not to be taken with others*. Students have two and a half hours to take each exam unless they have legally-allowed accommodations. Exams have HARD deadlines.

Final exam is worth 50 points, taken on Bb (Quizzes and Exams), and monitored by Respondus LdB; *it is not to be taken with others*. **The final exam cannot be retaken.** There will be questions regarding sustainability as well as cumulative chemistry content on the final exam. The final exam time allowed by the University is 2.5 hours unless a student has legally-allowed accommodations. **The final exam link will be available only during the 2.5 hour time-slot mandated by the University**. The final exam has a HARD deadline.

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Office of Disabilities (ODS) and Learning Support Services (LSS)

It is Villanova's policy to make reasonable accommodations for qualified individuals with disabilities. All students who require accommodations should access Clockwork[4] to complete the online intake. Please find registration guidelines and instructions at the LSS website[5] or the ODS website.[6] Contact the LSS Office at 610-519-5176 or the ODS office at 610-519-3209 for questions. IMPORTANT: *Please inform Dr. Katz of any academic or physical needs* ***in advance*** *of scheduled quiz/exam dates if special accommodations are required*, especially if the accommodations include extended testing. **If a student has legal accommodations that allow extended testing, this legally-allowed time will be automatically scheduled into the online tests, but only if Dr. Katz knows about them.**

Absences for Religious Holidays

Villanova University makes every reasonable effort to allow members of the community to observe their religious holidays, consistent with the University's obligations, responsibilities, and policies. Students who expect to miss a class or assignment due to the observance of a religious holiday should discuss the matter with Dr. Katz as soon as possible, preferably one week in advance. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the absence. For more information, please visit the website below.[7] Questions? Please ask Dr. Katz.

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[4] Register academic/physical accommodations at Clockwork: https://vuclockwrk.villanova.edu/clockwork/custom/misc/home.aspx
[5] LSS website http://learningsupportservices.villanova.edu
[6] ODS website http://learningsupportservices.villanova.edu
[7] Religious holiday policy https://www1.villanova.edu/villanova/provost/resources/student/policies/religiousholidays.html

Academic Integrity

All students are expected to uphold Villanova's Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action, per University rules. For the College's statement on Academic Integrity, please consult the Student Guide to Policies and Procedures document.[8] Please view the University's Academic Integrity Policy and Code, as well as other useful information, at the Academic Integrity Gateway web site.[9] In case there is any confusion about these facts, I'd like to state them clearly here: **Students may not work with, consult, or assist other students (or any person or online resource, including AI generators) during online quizzes and exams**.

Student exams are proctored by the Respondus LdB *so all students will be protected from accusations of cheating*. However, if instances of academic integrity arise during online testing (*which has unfortunately happened in the past)*, this will trigger all exams and quizzes to be moved offline. This would be disappointing, as **online exams/quizzes provide students with a high degree of flexibility and convenience**. Please *uphold the integrity of the online tests* so students may continue to enjoy this flexibility and convenience throughout the semester. Thanks!

Online Testing

Because a life can be so unpleasantly rigid, this course has some flexibility built into the (normally) most stressful part of the course. Quizzes and exams in this course will be taken on Bb and students will choose on what day they are taken from a range of days/times during which the quizzes/exams will be available. **Students must click the quiz/exam link within the times shown below in red/purple.**

To protect test integrity, the Respondus LdB with Monitor (*special browser that locks students out of all applications and prevents communication with others*) will be used to proctor all exams and quizzes. On the first day of class, students will be guided through the steps to download this browser. If there is an emergency during an exam or quiz, students can call/text Dr. Katz. No worries! The point of the LdB is that students sit, write their test, it gets submitted, and there's video proof of it. ***This is why students need a webcam.*** *It protects students from accusations of academic integrity violations.*

Links for the quizzes and exams (described below) can be found on the Quizzes and Exams page on Bb. For each quiz, students are given a 2-hour time-period and for exams, including the final exam, students are given a 2.5-hour time-period. Students who have legally allowed accommodations (***that are registered with LSS or ODS***; see above), the legally-allowed testing time will be scheduled directly into their quizzes and exams. Students will choose where they will sit, and what time and day their quiz or exam will begin **within red/purple time periods** listed below. **This is not the case for the final exam; there is only a two and a half hour window to take the final exam**, not a 5-day window. The final exam is also taken online, but the link is only available for 2.5 hours.

*The test links **disappear** after the **Monday** (or $) **5:00 pm** (or $$) times shown below:*	
Week 1/2 Quiz 1 – Topic: Ch 1	Thurs, Jan 16, 5:00 pm – Mon, Jan 20, 5:00 pm
Week 2/3 Quiz 2 – Topic: Ch 2	Thurs, Jan 23, 5:00 pm – Mon, Jan 27, 5:00 pm
Week 3/4 Quiz 3a – Topic: Ch 3 – part 1	***Tues***, Jan 28, 5:00 pm – ***Sat***,$ **Feb 1**, 5:00 pm*
Week 3/4 Quiz 3b – Topic: Ch 3 – part 2	***Fri*, Jan 31**, 5:00 pm – Mon, Feb 3, 5:00 pm*
**Two Ch 3 quizzes are staggered over one week*	
Week 4/5 Quiz 4a – Topic: Ch 4 – part 1	Thurs, Feb 6, 5:00 pm – Mon, Feb 10, 5:00 pm
Week 5/6 Exam 1 – Topic: Ch 1 through Ch 4.3	**Thurs, Feb 13, 5:00 pm – Mon, Feb 17, 5:00 pm**
Week 6/7 Quiz 4b – Topic: Ch 4 – part 2	*Thurs*, Feb 20, 5:00 pm – Mon, Feb 24, 5:00 pm
Week 7/8 Quiz 5 – Topic: Ch 5	*Wed*, Feb 26, 5:00 pm – Mon, Mar 10, 5:00 pm**
***Available before…through…and after break*	
Week 8/9 Quiz 6 – Topic: Ch 6	Thurs, Mar 13, 5:00 pm – Mon, Mar 17, 5:00 pm
Week 9/10 Exam 2 – Topic: Ch 4 – 6	**Thurs, Mar 20, 5:00 pm – Mon, Mar 24, 5:00 pm**
Week 10/11 Quiz 7 – Topic: Ch 7	Thurs, Mar 27, 5:00 pm – Mon, Mar 31, 5:00 pm
Week 11/12 Quiz 8 – Topic: Ch 8	Thurs, Apr 3, 5:00 pm – Mon, Apr 7, 5:00 pm
Week 12/13 Quiz 9 – Topic: Ch 9	Thurs, Apr 10, 5:00 pm – Mon, Apr 14, 5:00 pm
Week 13/14 *No testing during Easter Break*	
Week 14/15 Exam 3 – Topic: Ch 7 through Ch 10	Thurs, Apr 24, 5:00 pm – *Sun*,$ *Apr 27*, 5:00 pm
Week 15/16 – **no testing allowed this week** – Excellent time for retakes (and remediation) – schedule retakes by May 1, 2025.	
Final Exam – Ch 1 – 10 – Live Link: *Tuesday*,$ *May 6*, 2025, 4:00 pm – 6:30 pm$$	
Please do not schedule travel during our final exam time which is determined by the University, and I am not permitted to change.	

[8] Student Guide to Policies and Procedure document http://www1.villanova.edu/villanova/artsci/undergrad/enchiridion.html
[9] Academic Integrity Gateway https://library.villanova.edu/research/subject-guides/academicintegrity

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Learning Objectives – upon completion of this course, students will demonstrate knowledge of these topics:

- Matter and Measurement: dimensional analysis, SI units (metric system), temperature scales, significant figures, scientific notation, volume, density, atomic structure, isotopes, periodic table, atomic mass.
- Chemical Nomenclature: writing formulas of molecules and ions; covalent and ionic nomenclature, names of acids and hydrates; diatomic molecules; polyatomic ions.
- The Mole Concept: molar mass, percent composition; empirical and molecular formulas, combustion analysis.
- Chemical Reactions and Stoichiometry: balancing eqns, types of eqns, acid/base, redox, metathesis, solubility rules, precipitation rxns, oxidation states, net ionic equations, molarity, titration, theoretical yield, percent yield, limiting reagent, and solution stoichiometry.
- Gas Laws: pressure definitions and units; Boyle's Law, Charles' Law, combined and ideal gas laws, Avogadro's Law, Dalton's Law of partial pressures, Kinetic Molecular Theory of gases, and stoichiometry of gas reactions.
- Thermochemistry: definitions of energy, work, standard state, standard enthalpy of formation, and heat; thermochemical eqns, calorimetry, and standard enthalpy changes from heat of formation and Hess's Law.
- Electronic Structural Models and Periodicity: EM spectrum, Bohr's model, hydrogen atom spectrum; Hund's Rule, electron configuration, valence electrons and periodic trends: effective nuclear charge, ionization energy, and atomic radius.
- Bonding theories: ionic bonding, Lattice Energy, electronegativity, covalent bonding: octet rule and its exceptions, Lewis structures, formal charge, resonance, Valence Bond Theory, VSEPR, electronic and molecular geometry, dipole moments, hybridization, and bond enthalpy.
- Intermolecular Forces: ion-dipole, dipole-dipole, dispersion forces, hydrogen bonding; phase changes, heating curve, phase diagram, vapor pressure, surface tension, viscosity, and capillary action.
- Sustainability: domains of sustainability, hidden costs of EV, Green Chemistry, Ocean Acidification, Greenhouse gases, Water Energy Nexus, Solids in the Environment (Pb, Cr(IV), PFAS).

Expected Daily Class Schedule

The expected daily class schedule can be found below and on the Semester-at-a-glance page on Bb. This calendar gives details regarding the chapters to be covered during class, the HW assignments and due dates, the reading assignments, and the quiz and exam topics with due dates. The schedule may be updated periodically to best accommodate student learning needs. If the schedule changes, it will be announced it at the onset of class and posted it on Bb. Questions about the schedule or any update? Please ask Dr. Katz.

Expected Daily Class Schedule CHM 1151 for Spr 2025

This calendar may change based upon what's best for student learning needs

Week Mo Day	Topics to be covered during Tues classes (*W - CHM 1103 info – applies **only** to students co-enrolled in CHM 1103*) Topics to be covered during Thurs classes	**Weekly Assignments** Due 5:00 Friday (soft) **Achieve;** Chem Journal; Group Activities (Points)	Weekly Reading Assignments to be read *before class* **OSAF,** eBook (Achieve 2025) L2Ls due 11:59 pm (10)
1 Jan 14 16	T Co-create class/group expectations; find partners; download Respondus LdB; Syllabus 2.0 *W CHM 1103: No lab during first week;* to prepare for next week, go to CHM 1103 Bb page Th Problem solving; Ch 1 Dimensional analysis, significant figures, and Domains of Sustainability	• MUST "quiz" Achieve (5) • Ch 1 HW – Achieve (10) • Ch 1 – Bb Chem Journal (5) • Group Activity – Syllabus Scavenger Hunt (10) • Group Activity – Ch 1 Sustainability Activity (10)	Week 1 Reading **Before Thurs Class:** Read Ch 1: 1.4 – 1.6 **in OSAF**
Week 1/2 **Quiz Ch 1** – Live Link: Thurs Jan 16, 5:00 pm – Mon Jan 20, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
2 Jan 21 23	T Ch 2 Atomic Structure, Atomic mass, Empirical formula, The Mole, Molar mass. *W CHM 1103 Check in, Lab Intro Assignment, Sig Figs Assignment* Th Ch 3a EM Spectrum, wavelength, frequency, speed of light, Electron Configurations (3.4)	• Ch 2 HW – Achieve (10) • Ch 2 – Bb Chem Journal (5) • Electron Configuration self-guided lesson (10)	Week 2 Reading Tues: Read Ch 2.3 - 2.4. Thurs: Read: Ch 3.1 – 3.4 (electron configurations).
Week 2/3 **Quiz Ch 2** – Live Link: Thurs Jan 23, 5:00 pm – Mon Jan 27, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
3 Jan 28	T Ch 3b Periodic Trends: Z_{eff}, Covalent Radius, Cationic/Anionic Radius, IE, Shielding; ionic and molecular compounds; naming acids and diatomics	• Ch 3a HW – Achieve (10) • Ch 3b HW – Achieve (10) • Ch 3 – Bb Chem Journal (5)	Week 3 Reading Tues: Read Ch 3.5 – 3.7.

30	*W CHM 1103 Expt 1: Eyewash, Gloves, and Mass Balances* Th Compare/Contrast activity for Periodic Tables	• Group Activity – Periodic Table Activity (10)	L2L 1 (Syllabus 1.0) due Sun, Feb 2nd
Week 3/4 **Quiz Ch 3a** – Live Link: Tues Jan 28, 5:00 pm – Sat Feb 1, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
Week 3/4 **Quiz Ch 3b** – Live Link: Fri Jan 31, 5:00 pm – Mon Feb 3, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
4 Feb 4 6	T Ch 4a Covalent bonding; Electronegativity; Chemical nomenclature (*End Exam 1 Material*) *W CHM 1103 Expt 2: Accuracy and Precision of Laboratory Glassware, and Determining Density* Th Workshop day to Journal, work on HW etc.	• Ch 4a HW – Achieve (10) • Ch 4 – Bb Chem Journal (5)	Week 4 Reading Tues: Read 4.2 – 4.3 (**end Ex 1 material**) Thurs: Read 4.4 – 4.5
Week 4/5 **Quiz Ch 4a** – Live Link: Thurs Feb 6, 5:00 pm – Mon Feb 10, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
5 Feb 11 13	T Ch 4b Lewis structures, Formal charge, Resonance *W CHM 1103 Expt 3: Empirical Formula* Th **Review for Exam 1 – In class Practice Exam**	No deliverables during the week of an exam	Week 5 Reading Tues: Read Ch 4.6
Week 5/6 **EXAM – Ch 1 through Ch 4.3 (Naming, but no Lewis Structures)** Live Link: Thurs Feb 13, 5:00 pm – Mon Feb 17, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
6 Feb 18 20	T Ch 4b Molecular Structure and Polarity *W CHM 1103 Expt 4: Atomic Emission Spectroscopy* Th Ch 10.1 Intermolecular forces (IMFs); Ocean Acidification Activity (this material is on Quiz)	• Ch 4b HW – Achieve (10) • Group Activity: Ocean Acidification (10) • 10.1 HW/Quiz done w Ch 10 • *Exam 1 remediation due Monday next week (opt'l)*	Week 6 Reading Tues: Read Ch 4.6 Thurs: Read Ch 10.1 L2L 2 (Syllabus 1.0) due Sun, Feb 23rd
Week 6/7 **Quiz Ch 4b** – Live Link: Thurs Feb 20, 5:00 pm – Mon Feb 24, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
7 Feb 25 27	T Ch 5 VB Theory; Hybrid Orbitals; MOTheory *W CHM 1103 Expt 5: VSEPR and Polarity* Th Workshop day: Journal, work on HW, take **Quiz**	• Ch 5 HW – Achieve (10) • Ch 5 – Bb Chem Journal (5)	Week 7 Reading Tues: Read Ch 5.1 – 5.4
Week 7/8 **Quiz Ch 5** – Live Link: **WEDNESDAY** Feb 26, 5:00 pm – Mon Mar 10, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
Semester Recess March 3 – Mar 7			
8 Mar 11 13	T Ch 6 Formula Mass, Empirical Formula, Percent composition, Molarity, Dilution *W CHM 1103 Expt 6: Computational Chemistry* Th Workshop day to Journal, work on HW etc.	• Ch 6 HW – Achieve (10) • Ch 6 – Bb Chem Journal (5)	Week 8 Reading Tues: Read Ch 6.1 – 6.3 (end Ex 2 material) L2L 3 (Syllabus 1.0) due Sun, Apr 16th
Week 8/9 **Quiz Ch 6** – Live Link: Thurs Mar 13, 5:00 pm – Mon Mar 17, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
9 Mar 18 20	T Ch 7 Writing and Balancing chemical equations, Classifying Chemical reactions, Net ionic equations *W CHM 1103 Expt 7: Chemical Reactions* Th **Review for Exam 2: In-class Practice Exam**	No deliverables during the week of an exam	Week 9 Reading Tues: Read Ch 7.1 – 7.2
Week 9/10 **EXAM 2 – Ch 4 - 6** – Live Link: Thurs Mar 20, 5:00 pm – Mon Mar 24, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
10 Mar 25 27	T Ch 7 Reaction Stoichiometry, Reaction yields, Reaction Efficiency (Atom Economy) *W CHM 1103 Expt 8: Acid Base Titration* Th Ch 8 Gas Pressure, KMT, Gas Laws	• Ch 7 HW – Achieve (10) • Ch 7 – Bb Chem Journal (5) • *Exam 2 remediation due Monday next week (opt'l)*	Week 10 Reading Tues Read Ch 7.3 – 7.4 (How Sciences Interconnect box) Thurs: Read Ch 8.1 - 8.2, 8.5
Week 10/11 **Quiz Ch 7** – Live Link: Thurs Mar 27, 5:00 pm – Mon Mar 31, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
11 Apr 1 3	T Ch 8 Gas Stoichiometry, Dalton's Law *W CHM 1103 Expt 9: Gas Laws* W/X Apr 2 Th Ch 8 Greenhouse Gases and our Warming Climate; Workshop day to Journal, work on HW.	• Ch 8 HW – Achieve (10) • Ch 8 – Bb Chem Journal (5) • GHG Group Activity (10)	Week 11 Reading Tues: Read Ch 8.3 L2L 4 (Syllabus 1.0) due Sun, Apr 6th
Week 11/12 **Quiz Ch 8** – Live Link: Thurs Apr 3, 5:00 pm – Mon Apr 7, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
12 Apr	T Ch 9 Energy Basics and Calorimetry *W CHM 1103 Exp 10: Thermo and Calorimetry*	• Ch 9 HW – Achieve (10) • Ch 9 – Bb Chem Journal (5)	Week 12 Reading Tues: Read Ch 9.1 – 9.2

8 10	Th Ch 9 Enthalpy; Hess's Law, Bond Energy; Water Energy Nexus	• Water Energy Nexus Group Activity (10)	Thurs: Read Ch 9.3 – 9.4 Omit Lattice Energy and Born Haber Cycle Section 9.4
Week 12/13 **Quiz Ch 9** – Live Link: Thurs Apr 10, 5:00 pm – Mon Apr 14, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
13 Apr 15 17	T Ch 10 Properties of liquids and solids; Solids in the Environment Group Activity *W CHM 1103 NO LAB THIS WEEK* Th **NO CLASS – Easter Recess**	• Ch 10 HW – Achieve (10) • Ch 10 – Bb Chem Journal (5) • Solids in the Environment Group Activity (10)	Week 13 Reading Tues: Ch 10.2
NO scheduled testing during Easter Recess (Apr 16 – Apr 21)			
14 Apr 22 24	T Ch 10 Phase Diagrams and Heating curves *W CHM 1103 CATS, Technique test, Check out* Th **Review for Exam 3: In-class Practice Exam**	No deliverables during the week of an exam	Week 14 Reading Tues: Read Ch 10.3 – 10.4 L2L 5 (Syl 1.0) due Tues, Apr 29th
Week 14/15 **EXAM 3 – Ch 7 - 10** – Live Link: Thurs Apr 24, 5:00 pm – Mon Apr 28, 5:00 pm **(HARD)** **(Quizzes and Exams)**			
15 May 1	*T no class DEEMED A FRIDAY BY THE UNIVERSITY* *W CHM 1103 no lab meeting this week* Th **Final Exam Review;** Workshop time to finish past due work.	*Exam 3 remediation* (*opt'l*) and all past-due work to be submitted for grading due by **Thurs, May 1, 4:40 pm**	No Class on Tuesday April 29th
Final Exam – Ch 1 - 10 – Live Link Tuesday, May 6, 2025, 4:00 pm – 6:30 pm (HARD) **(Quizzes and Exams)**			

Grading System*

Faculty members are responsible for maintaining the integrity of the evaluation and grading system. Presented below is the Undergraduate Grading System:

Grading System developed by Villanova University modified to include numerical values by Dr. Stephanie Katz	
A ≥ 94 A- 90 – 93 B+ 87 – 89	is the highest academic grade possible; an honor grade, which is not automatically given to a student who ranks highest in the course but is reserved for accomplishment that is truly distinctive and demonstrably outstanding. It represents a superior mastery of course material and is a grade that demands a very high degree of understanding as well as originality or creativity as appropriate to the nature of the course. The grade indicates that the student works independently with unusual effectiveness and often takes the initiative in seeking new knowledge outside the formal confines of the course.
B 84 – 86 B- 80 – 83 C+ 76 – 79	is a grade that denotes achievement considerably above acceptable standards. Good mastery of course material is evident and student performance demonstrates a high degree of originality, creativity, or both. The grade indicates that the student works well independently and often demonstrates initiative. Analysis, synthesis, and critical expression, oral or written, are considerably above average.
C 70 – 75 C- 66 – 69 D+ 64 – 65	indicates a satisfactory degree of attainment and is the acceptable standard for graduation from college. It is the grade that may be expected of a student of average ability who gives to the work a reasonable amount of time and effort. This grade implies familiarity with the content of the course and acceptable mastery of course material; it implies that the student displays some evidence of originality and/or creativity, works independently at an acceptable level and completes all requirements in the course.
D 60 – 63 D- 57 – 59	denotes a limited understanding of the subject matter, meeting only the minimum requirement for passing the course. It signifies work that in quality and/or quantity falls below the average acceptable standard for passing the course. Performance is deficient in analysis, synthesis, and critical expression; there is little evidence of originality, creativity, or both.
F ≤ 56	indicates inadequate or unsatisfactory attainment, serious deficiency in understanding of course material, and/or failure to complete requirements of the course.
N	Incomplete: course work not completed.

*The original grid (without numbers) was obtained from: https://www1.villanova.edu/villanova/provost/resources/student/policies/grades.html

Please sign (next page) either Syllabus 1.0 or Syllabus 2.0 before January 31st. **Upload the signed syllabus page** on the Syllabus and Professor page on Bb. (See below for explanation). Thanks!

CHM 1151 | General Chemistry I – Syllabus 1.0 | Spring 2025
Section 001 | Department of Chemistry and Biochemistry | CRN 35967

Signing the Syllabus

Each student must decide whether they want to use Syllabus 1.0 or Syllabus 2.0. The breakdown for each syllabus is shown below:

Graded Assessments	Syllabus 1.0		Syllabus 2.0		Deadline
Link2Learn (L2L) Assignment	7.69 %	50 points	----	-----	soft
Participation/attendance	5.38 %	35 points	35 points	5.83 %	-----
Homework (10 pts each)	18.46 %	120 points	120 points	20.00 %	soft
Sustainability (10 points each)	12.31 %	80 points	80 points	13.33 %	soft
Chemistry journal (5 points each)	7.69 %	50 points	50 points	8.34 %	soft
Online quizzes (10 points each)	17.69 %	115 points	115 points	19.17 %	HARD
Online exams (*3 @ 50 pts each*)	23.07 %	150 points	150 points	25.00 %	HARD
Online final exam (50 points)	7.69 %	50 points	50 points	8.34 %	HARD
611 points (94%) needed for an A	~100 %	650 points	600 points	564 points needed for an A	

Link2Learn: 5 times, for 10 pts each, write (a) the *title* of a science- or sustainability-related or adjacent article (1 pt); (b) where it was found it (1 pt) (this is not the citation; I'm curious where/how students find news – Tik Tok, X, Google); (c) "Decolonize the literature" (2 pts) – i.e., find the author's home country, country/city of current residence, gender, race and/or ethnicity, and education; (d) write a brief summary of the article (3 pts) – that is, summarize what was learned it in approximately 1 – 2 paragraphs; and (e) provide a complete MLA/APA-type citation. ***Bonus points*** may be earned if what is shared is so interesting that Dr. Katz feels compelled to share it with the class. If all five L2Ls are submitted on time, an extra 5 bonus points will be applied to L2L 5.

Link2Learn due dates - at 11:59 pm

- Link2Learn 1 due February 2, 2025 (Sun)
- Link2Learn 2 due February 23, 2025 (Sun)
- Link2Learn 3 due March 16, 2025 (Sun)
- Link2Learn 4 due April 6, 2025 (Sun)
- Link2Learn 5 due April 29, 2025 (Tues)

Please choose between Syllabus 1.0 and Syllabus 2.0, sign below, and upload on Bb (on Syllabus and Professor page) by January 31st

Semester Contract – A signature required for every student in this course...

Students who want **to complete L2L assignments**, and agree to a **650-point semester**, please fill out the top form:

I, (PRINT/type student NAME HERE): ____, am providing my signature below,

Student signature: __________________________________ (e-signature is allowed) to indicate that *I choose Syllabus 1.0* (meaning *I agree to complete the **L2L*** assignments) for CHM 1151-001 during Spring 2025 with Dr. Katz.

If you signed above, please do not sign below. If you plan to sign below, please do not sign above. Thank You.

✦ ✦

Students who **do NOT want to complete L2L assignments**, and agree to a **600-point semester**, please fill out this one:

I, (PRINT/type student NAME HERE): ____, am providing my signature below,

(Student SIGN HERE): ________________________________ (e-signature is allowed) to indicate that *I choose Syllabus 2.0* (with *NO required **L2L*** assignments) for CHM 1151-001 during Spring 2025 with Dr. Katz.

✦ ✦ Please sign only one above and UPLOAD on the Syllabus and Professor page ✦ ✦